

An aphid-resistant plant metabolite as a candidate aphicide: Insight into the bioactivity and action mode of betulin against aphids

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eLife Assessment

This study presents **important** findings on the insecticidal mechanism of betulin, a plant-derived metabolite, in controlling the aphid *Myzus persicae* and it provides a demonstration that betulin targets the GABA receptor in aphids, with strong supporting evidence from transcriptomic, biochemical, electrophysiological, and genetic approaches. In particular, the identification of a specific conserved residue (THR228) critical for betulin binding advances our understanding of insect neuropharmacology and offers translational potential for pest management strategies. The evidence supporting the primary claims is **solid**, with well-integrated methodologies and appropriate controls; however, some interpretative and methodological limitations remain, including the option to further explore off-target effects, as well as the broader evolutionary and ecological context. Addressing these points would strengthen the broader implications of the study.

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Abstract

Pest-resistant plants usually utilize secondary metabolites to cope with insect infestation. Betulin, a key bioactive compound in aphid-resistant wild peach, possesses promising applications in crop protection. Here, betulin, in both greenhouse and field experiments, displayed excellent control efficacy against *Myzus persicae*. RNA-seq, qRT-PCR, and western blotting revealed that betulin significantly inhibited the expression of *MpGABR* (encoding a GABA_A receptor). Besides, RNAi-mediated silencing of *MpGABR* markedly increased aphid sensitivity to betulin. Furthermore, MST and voltage-clamp assays indicated that betulin bound to *MpGABR* ($K_d = 2.24 \mu\text{M}$) and acted as an inhibitor of *MpGABR*. Molecular docking, mutagenesis and genome editing suggested that THR228 is a critical and highly conserved site

in MpGABR that betulin binds to specifically, causing aphid death. Overall, the activity of betulin depends on specific targeting and inhibition of MpGABR. Elucidating the mechanism of action of this peach-derived insecticide may offer a sustainable green strategy for aphid control.

1. Introduction

Over more than 400 million years of coevolution with herbivores, plants have developed a variety of morphological,¹ biochemical,² and molecular³ defense strategies against insect herbivore attack. Plant secondary metabolites are highly valuable natural compounds, offering abundant and potent biochemical defense driven by plant-insect coevolution.⁴ Interestingly, wild relatives of crops with insect resistance usually accumulate large amounts of biologically active compounds.^{5,6} Accordingly, investigating the bioactivities and potential mechanisms of action of compounds derived from wild resistant germplasms may provide effective and sustainable strategies for pest control in crop protection.

Myzus persicae (Sülzer) is one of the most destructive sap-feeding pests worldwide. It is able to settle on more than 400 plant species globally, displaying extraordinary polyphagia and a wide range of hazards.⁷ In addition to ingesting phloem saps and secreting honeydew, *M. persicae* can transmit various plant viruses,⁸ resulting in severe losses in crop yields. To date, the principal control strategy for *M. persicae* has relied on synthetic insecticides, including pyrethroids, organophosphorus compounds, carbamates and neonicotinoids.^{9–11} However, long-term irrational application of these chemical insecticides has led to environmental pollution and promoted the resistance of aphids to insecticides.^{11–13} Therefore, it is necessary to develop novel green insecticides as potential alternatives to synthetic insecticides. Fortunately, plant secondary metabolites are a “treasure trove of active compounds” that serve as a rich resource for the development of promising green insecticides.

Our previous study revealed that *Prunus davidiana*, a close wild relative of cultivated peach, displays strong resistance to *M. persicae* through the accumulation of high levels of betulin.^{3,5} In addition, betulin, a lupane-type triterpene, possesses strong insecticidal activity and is a promising substance for the development of novel aphid-control insecticides. A number of studies have reported that betulin and its derivatives exhibit a wide range of pharmacological activities,¹⁴ which have been used to treat various diseases, such as cancer,¹⁵ glioblastoma,¹⁶ inflammation¹⁷ and hyperlipidemia.¹⁸ However, the insecticidal mechanism of betulin against aphids remains unclear.

Gamma-aminobutyric acid (GABA) receptors have been confirmed to be targets of terpenoids that impair neuronal function in insect herbivores.¹⁹ In mammals, there are two types of GABA receptors: ionotropic (GABA_A) and metabotropic (GABA_B) receptors. GABA_A receptors (GABRs) are heteropentameric ligand-gated ion channels in the central nervous system that conduct chloride and bicarbonate ions and are the target of numerous drugs for the treatment of neuropsychiatric disorders.²⁰ GABRs are composed of 5 different types of subunits, each with 4 helical transmembrane domains (TM1-TM4), of which TM2 is located at the center of the pentamer and forms an ion channel.²¹ However, the structure and assembly of insect GABRs have not been fully elucidated. The first GABA receptor subunit identified in insects is encoded by *Rdl*, which confers resistance to dieldrin in *Drosophila melanogaster*.²² Although a variety of terpenoids play roles as positive allosteric modulators or noncompetitive antagonists at GABRs,¹⁹ whether betulin targets the GABR in *M. persicae* needs to be further investigated.

In this study, betulin was confirmed to have excellent control efficacy against *M. persicae* in both greenhouse and field experiments. RNA-seq, qRT-PCR, and western blotting assays revealed that betulin significantly inhibited the expression of *MpGABR* in aphids. In addition, RNAi-mediated

silencing of *MpGABR* markedly increased aphid sensitivity to betulin. Furthermore, microscale thermophoresis (MST) and voltage-clamp assays indicated that betulin can bind to *MpGABR* ($K_d = 2.24 \mu\text{M}$) and act as an inhibitor ($\text{EC}_{50} = 20.66 \mu\text{M}$) of *MpGABR*. Molecular docking analysis suggested that betulin bound to *MpGABR* via the THR228 amino acid residue. This site is highly conserved in the family Hemiptera and may be a critical specific binding site for betulin in *MpGABR* in aphids. Mutagenesis and genome editing experiments revealed that betulin bound specifically to this amino acid residue in aphids but not in *Drosophila*, resulting in aphid death. This study elucidated the insecticidal mechanism of betulin involving the targeting of *MpGABR*, providing a sustainable green strategy for aphid control.

2. Materials and methods

2.1 Test insects

M. persicae populations were collected from tobacco (*Nicotiana tabacum* L.) leaves. Wild-type (WT) flies (*D. melanogaster*) were purchased from Fungene Biotechnology Co., Ltd., Jiangsu, China. The test insects were cultured in a greenhouse (25 °C, 70% relative humidity and 14 h light/10 h dark cycle) in Beibei, Chongqing, China.

2.2 Bioassay

The toxicity of betulin against *M. persicae* was measured using a slip-dip bioassay as previously described.⁵ Thirty apterous adult *M. persicae* individuals were gently transferred on tobacco (*N. tabacum*) leaves using a fine brush. The tobacco leaves were subsequently immersed in 0.1% (v/v) Tween-80 and 3% (v/v) acetone solution containing betulin or pymetrozine (as a positive control) at 6 doses (0, 0.0625, 0.125, 0.25, 0.5, 1, and 2 mg·mL⁻¹) for 5 seconds. After drying, the leaves with *M. persicae* were placed on wet filter paper and cultured on a water-soaked sponge for 48 h. Three replicates were tested for each treatment. Furthermore, the LC₃₀, LC₅₀ and LC₇₀ values of betulin were calculated using log-probit analysis via IBM SPSS Statistics (v.22.0, Chicago, IL, USA).²

2.3 Control effect of betulin against *M. persicae*

The control effects of betulin against *M. persicae* in the greenhouse and field were evaluated by a spraying-based method as described previously.²³ The commercial insecticide pymetrozine and an aqueous solution containing 3% (v/v) acetone and 0.1% (v/v) Tween-80 were employed as positive and negative controls, respectively. Sprays of 0.1641 mg·mL⁻¹ (LC₅₀) betulin and 1.0612 mg·mL⁻¹ (LC₅₀) pymetrozine (Table S1) were prepared in aqueous solutions including 3% (v/v) acetone and 0.1% (v/v) Tween-80, respectively. Before the control effect assays, each tobacco (*N. tabacum* L.) seedling was pre-inoculated with 60 apterous adults of *M. persicae*. In the greenhouse trial, each treatment included 8 replications, and each replication consisted of 3 tobacco plants growing under controlled conditions as described above. The field trial (Beibei District, Chongqing, China.) was conducted under natural conditions with an average temperature of 20 °C. Each treatment had 8 replications, covering an area of 20 m², with 3 protected rows per replication. Furthermore, each tobacco plant was sprayed with 20 mL of the corresponding solution with an electric sprayer. Mortality was recorded after 1, 5, 9, and 14 days of treatment. The control efficacy of betulin against *M. persicae* was calculated according to the methods of Zhou.²³

2.4 RNA-seq assay

After exposure to 0.1641 mg·mL⁻¹ (48 h LC₅₀) betulin for 48 h, 100 wingless *M. persicae* adults were sampled, and total RNA was extracted via TRIzol reagent (Tiangen, Beijing, China) to construct mRNA sequencing libraries. RNA-seq analysis was performed on the Ion Proton BGISEQ-500 platform (Beijing Genomic Institute, BGI, China). After the raw reads were filtered, the clean

reads were aligned to the *M. persicae* reference genome, which was downloaded from the National Center for Biotechnology Information database, using HISAT2 (v2.1.0).²⁴ The expression levels of genes in each sample were normalized to the fragments per kilobase million mapped reads (FPKM) value. The differentially expressed genes (DEGs) between groups were identified using the fold change factor ($|\text{Log}_2(\text{FC})| > 1$). Functional enrichment analysis of DEGs was performed using the Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) databases.

2.5 Validation by quantitative real-time PCR (qRT-PCR)

The gene-specific primers used for qRT-PCR, which were designed with Primer 5.0, are listed in Table S2. According to the manufacturer's specifications for the Bio-Rad iQTM SYBR[®] Green Supermix Kit, qRT-PCR was performed with the following conditions: 60 °C for 15 s, followed by 48 cycles of 60 °C for 30 s, and 72 °C for 30 s. The *ribosomal protein S18* gene (*RPS18*) was used as the reference gene to normalize the expression levels. The relative expression levels of the target genes were calculated using the $2^{-\Delta\Delta C_t}$ approach.²⁵ Three biological replicates were used for each assay.

2.6 Cloning, bioinformatics and phylogenetic analysis

The cDNA of *M. persicae* was synthesized from the extracted total RNA using the PrimeScript[®] 1st Strand cDNA Synthesis Kit (TaKaRa, Japan). The coding sequences of *MpGABR* (encoding GABA_A receptor), *MpGABRAP* (encoding GABA_A receptor-associated protein) and *MpGABRB* (encoding GABA_A receptor β subunit) were cloned from *M. persicae* cDNA using the corresponding specific primers (Table S2) and then inserted separately into the pGEM-T Easy vector. The recombinant plasmids were subsequently transferred into *Escherichia coli* DH5 α competent cells for sequencing. BioXM (v2.7.1) was used to analyze the sequence information of these three genes, including the length, molecular weight, and isoelectric point of the deduced amino acid sequence. MEGA7 was used for multiple amino acid sequence alignment (ClustalW) and phylogenetic analysis (neighbor-joining method with 1000 replicates) of *MpGABR*.

2.5 In vitro RNAi assay

To knock down the target genes (*MpGABR*, *MpGABRAP*, and *MpGABRB*), *MpGABR*-, *MpGABRAP*-, and *MpGABRB*-dsRNA were artificially synthesized using the dsRNA Synthesis Kit (Thermo Scientific, Vilnius, Lithuania, EU). In accordance with a previously described method,²⁵ fresh tobacco leaves were cut into 3.0 cm-diameter discs and placed in an oven at 50 °C for 3 min. Subsequently, the dried leaf discs were exposed to 10 μL of diethyl pyrocarbonate-treated nuclease-free water (DEPC-water), dsGFP (green fluorescent protein, 1000 ng $\cdot \mu\text{L}^{-1}$, negative control), and *MpGABR*-, *MpGABRAP*-, or *MpGABRB*-dsRNA (1000 ng $\cdot \mu\text{L}^{-1}$) at 25 °C for 5 h. After the leaf discs had absorbed the solution, they were placed on wet filter paper on a water-soaked sponge. Subsequently, 30 wingless adult *M. persicae*, starved for 24 h, were fed on the treated leaf discs for 48 h. The surviving aphids were collected after exposure for qRT-PCR.

2.6 Prokaryotic expression of *MpGABR* and western blotting

To produce the *MpGABR* protein *in vitro*, an *Escherichia coli* expression system was constructed.²⁶ Briefly, the coding sequence of *MpGABR* was inserted into the expression vector PET-B2M and transformed into BL21 (*DE3*) competent cells. To induce the expression of the *MpGABR* protein, 2 mM isopropyl β -D-1-thiogalactopyranoside (IPTG) was used, and then, the transformed BL21 cells were incubated at 180 rpm and 30 °C for 24 h. Finally, the protein was extracted and purified using Solarbio GST-tag Purification Resin for subsequent assays.

The purified *MpGABR* protein resolved on a 15% SDS-PAGE (sodium dodecyl sulfate-polyacrylamide gel electrophoresis) gel and then transferred to a polyvinylidene fluoride (PVDF) membrane at 20 V for 1 h. Moreover, 1 $\times$ TBST (Tris-Borate-Sodium Tween-20) containing 5% fat-free milk powder was used to block the PVDF membrane. Then, the PVDF membrane was

incubated with an antibody against MpGABR (polyclonal rabbit antibody, provided by GeneCreate Biotechnology) at 25 °C for 1 h. Subsequently, the membrane was further incubated with the corresponding horseradish peroxidase (HRP)-conjugated goat anti-rabbit immunoglobulin G (IgG) antibody at 25 °C for 1 h and then washed three times with TBST. Finally, the bands on the membrane were visualized using a Chemi Doc MP Imaging System (Bio-Rad) after incubation with a chemiluminescence reagent (ECL, Bio-Rad).

2.7 Molecular docking

In accordance with a previously reported method,²³ the protein structure and function of MpGABR were predicted via the I-TASSER server (Iterative Threading ASSEmbly Refinement). AutoDockTools software (v1.5.7) was used to conduct molecular docking. Furthermore, PyMOL (v3.1) and Discovery Studio (v4.5) were used to evaluate the molecular docking model.

2.8 Site-directed mutagenesis

The cDNA of *MpGABR* was inserted into the PET-B2M vector and used as the template for site-directed mutagenesis. In accordance with the manufacturer's specifications, the Fast MultiSite Mutagenesis Kit (TransGen, Beijing, China) was used to generate the multisite mutations in MpGABR.²⁷ Then, the MpGABR mutants, with the mutations R224A (ARG224 replaced by ALA), A226T (ALA226 replaced by THR), F227Y (PHE227 replaced by TYR), and T228R (THR228 replaced by ARG), were amplified, and the amplicons were used to replace the *Bam*HI-*Sma*I fragment in the recombinant MpGABR plasmid.

2.9 Method for *Drosophila* genome editing

To generate *Drosophila* mutants bearing the DmGABR^{R122T} mutation, homologous to THR228 in MpGABR, the Clustered Regularly Interspaced Short Palindromic Repeats-Cas9 (CRISPR/Cas9) genomic editing strategy was used. First, CRISPR Optimal Target Finder (<http://targetfinder.flycrispr.neuro.brown.edu/>) was used to generate short guide RNA (sgRNA) targets without off-target hits. Then, the Precision gRNA Synthesis Kit (Thermo Scientific) was used to synthesize the sgRNA. The 228 bp homologous arm in R122T, for homology-directed repair, was subsequently employed to synthesize single-stranded DNA (ssDNA) bearing the desired mutation (DmGABR^{R122T}). Subsequently, 1 nL of a mixture containing 160 ng·μL⁻¹ ssDNA, 300 ng·μL⁻¹ both sgRNAs, or 90 ng·μL⁻¹ Cas9 protein was injected into *Drosophila* embryos via the Drummond Nanoject III (Broomall, PA, USA) system. CRISPR/Cas9-induced mutations were verified by amplifying and sequencing gDNA flanking the genome editing target site. Finally, according to the traditional topical application and bioassay approach described previously,²³ the acute insect toxicity of betulin in genome-modified or nonmodified (WT) 3-day-old adult flies was determined. The oligonucleotide sequences used for PCR, the sgRNA synthesis template and the fragment of donor DNA used for homology-directed repair (HDR) are listed in Table S3.

2.10 Microscale thermophoresis (MST) assay

The binding affinity of MpGABR for betulin was measured via a NanoTemper Technologies Monolith Kit (MO, Munich, Germany) according to the manufacturer's specifications. Briefly, MpGABR was fluorescently labeled via a protein labeling kit (RED-NHS 2nd generation, MonolithTM) and then mixed with sixteen concentrations (from 5 nM to 60 mM) of betulin. The mixtures were subsequently loaded into NanoTemper capillaries for MST measurements with 40% LED and 60% MST power at 25 °C. MO Control NanoTemper Technologies GmbH software (v2.3) and MO Affinity Analysis software (v2.3, NanoTemper Technologies GmbH) were used to acquire and analyze the recorded data, respectively. The dissociation constant (K_d), indicating the binding affinity, was calculated using data from three replicates for each experiment.

2.11 Electrophysiological recording

In accordance with previously described methods,²⁸ oocytes isolated from mature healthy *Xenopus* females were digested at 25 °C for 1 h to remove the follicular layer. Individual oocytes were selected to express *MpGABR* and incubated at 18 °C with 1× Ringer's solution (96 mM NaCl, 5 mM MgCl₂, 5 mM HEPES, 2 mM KCl, and 0.8 mM CaCl₂) supplemented with 550 mg·mL⁻¹ sodium pyruvate, 100 mg·mL⁻¹ streptomycin, 50 mg·mL⁻¹ tetracycline, and 5% dialyzed horse serum. After 3 days of incubation, the whole-cell currents were recorded via a two-electrode voltage-clamp. The data were collected and analyzed using Digidata 1440A and pCLAMP 10.2 software (Axon Instruments Inc., Union City, CA, USA), respectively.²⁹

2.12 Statistical analysis

Statistical analysis was performed using SPSS (v22). Statistical significance was determined using one-way analysis of variance (ANOVA) for differences among groups. The replicate numbers are indicated by the black dots in the figures.

3. Results

3.1 Control efficacy of betulin against aphids

Bioassays revealed that the LC₅₀ values of betulin and pymetrozine against *M. persicae* at 48 h were 0.1641 and 1.0612 mg·mL⁻¹, respectively (Table S1). To assess the control efficacy of betulin, tobacco plants infected with *M. persicae* were exposed to betulin (0.1641 mg·mL⁻¹) and pymetrozine (1.0612 mg·mL⁻¹, as a positive control) in both the greenhouse and field. In the greenhouse test (**Fig. 1A-D**), after 14 days of treatment, *M. persicae* in the CK group (negative control) reproduced rapidly, and tobacco leaves turned yellow and wrinkled, whereas tobacco leaves in the betulin and pymetrozine treatment groups were barely affected by the aphids (**Fig. 1A**). The control efficacy of betulin against *M. persicae* was significantly greater than that of pymetrozine after 1 day of treatment, while there was no significant difference between the control efficacies of betulin and pymetrozine after 5, 9, and 14 days of treatment (**Fig. 1E**). After 14 days of treatment, the control efficacies of betulin and pymetrozine reached 91.54% and 95.15%, respectively. Throughout the field test (**Fig. 1F**), no difference in the control efficacy was observed between betulin and pymetrozine, with values of 87.79% and 90.14%, respectively, after 14 days of treatment (**Fig. 1G**). These results suggest that betulin has immense potential for development as a commercial aphid insecticide like pymetrozine.

3.2 Expression patterns of genes in aphids after treatment with betulin

To investigate the expression patterns of genes in *M. persicae* after exposure to betulin, RNA-seq was performed on *M. persicae* with 0.1641 mg·mL⁻¹ betulin treatment for 48 h and without (CK) (**Fig. 2A**). Principal coordinate analysis (PCA) revealed that the expressed genes were strongly clustered in the CK group and betulin group (**Fig. 2B**). Compared with those in the CK group, there were 130 up- and 41 downregulated significant differentially expressed genes (DEGs) in *M. persicae* after betulin treatment (**Fig. 2C and D**, Table S4). The validation by qRT-PCR revealed that the relative expression trends of the fifteen DEGs determined via qRT-PCR were similar to those detected via RNA-seq, supporting the reliability of the RNA-seq data (**Fig. 2E**). To further analyze the functions of the DEGs, GO (Gene Ontology) and KEGG (Kyoto Encyclopedia of Genes and Genomes) enrichment analyses were performed. GO term enrichment demonstrated that the DEGs were enriched mainly in the GABA_A receptor activity term (**Fig. 2F**). Interestingly, KEGG enrichment also found that the DEGs were enriched in the GABAergic signaling related pathway

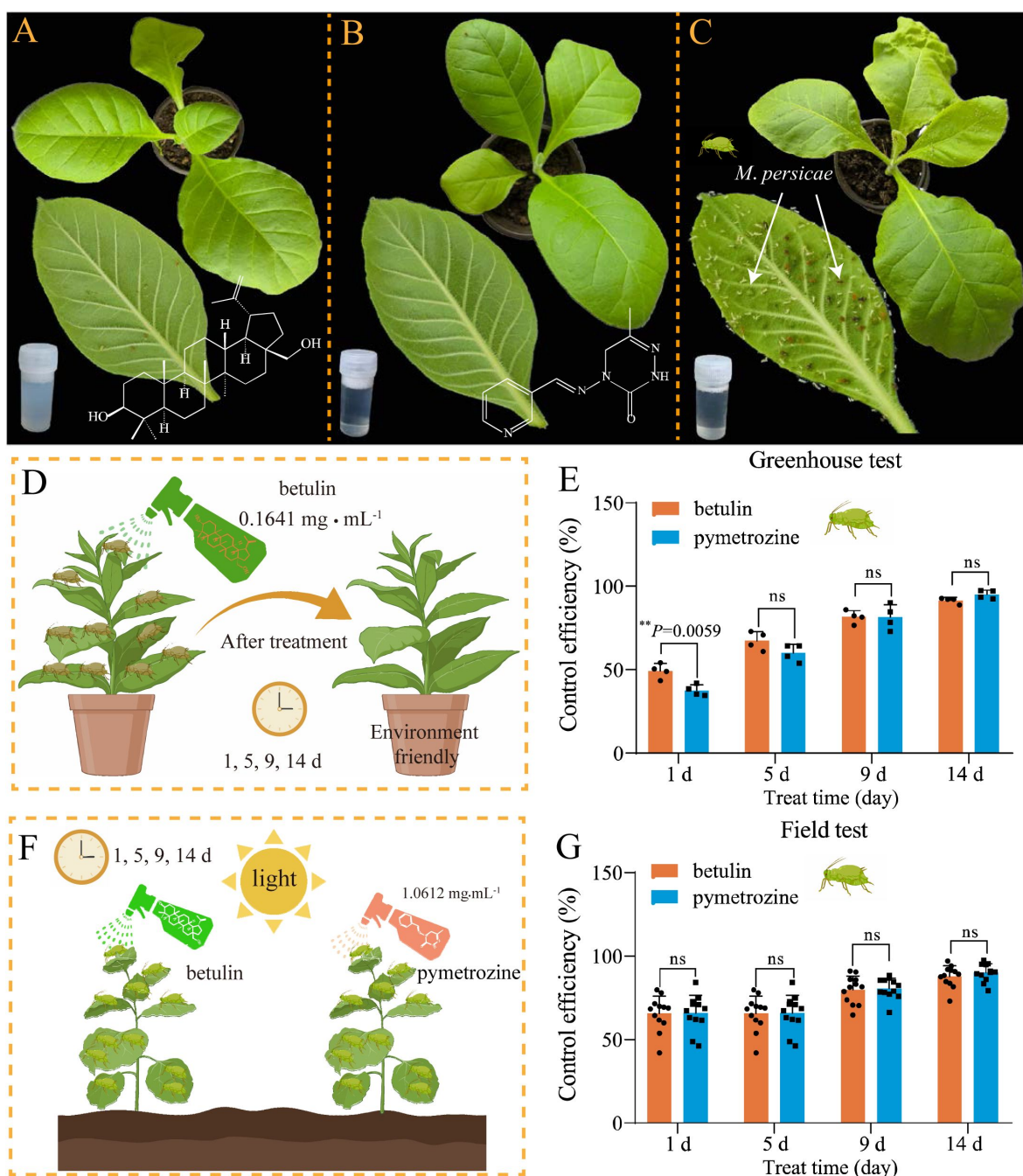


Figure 1.

The aphicidal activity of plant-derived derivative betulin against *M. persicae*.

(A-G) Representative images of the control effects of betulin (A, 0.1641 mg·mL⁻¹, 48 h LC₅₀ value of betulin), pymetrozine (B, 1.0612 mg·mL⁻¹, a positive control), and CK (C, a negative control) against *M. persicae* in greenhouse (A-E) and field (F, G) tests after treatment for 14 d. The data are shown as the mean ± SD from twelve independent experiments. ***p* < 0.01. ns, not significant.

and GABAergic synapse terms (**Fig. 2G**). Additionally, the expression of DEGs related to GABR, including *MpGABR* (encoding GABA_A receptor), *MpGABRAP* (encoding GABA_A receptor-associated protein) and *MpGABRB* (encoding GABA_A receptor β subunit), in the betulin group was significantly lower than that in the CK group (**Fig. 2H**). In particular, the $\log_2$ (fold change) value of *MpGABR* was the lowest, at -3.8 (Table S4). These results suggested that *MpGABR* may be a candidate target for betulin against *M. persicae*.

3.3 *MpGABR* expression was inhibited by betulin

Analysis of the sequence information revealed that the *MpGABR*, *MpGABRAP* and *MpGABRB* proteins have 694, 118, and 250 amino acids, respectively, with calculated molecular weights of 77.20, 14.08, and 28.01 kDa and isoelectric points of 9.85, 9.63, and 7.20, respectively (Table S5). Additionally, structure prediction indicated that *MpGABR*, *MpGABRAP* and *MpGABRB* contain 4, 0, and 1 transmembrane helical domain, respectively (**Fig. 3A**), implying that only *MpGABR* has a complete transmembrane structure.

To further verify the inhibitory effect of betulin on GABR-related gene expression, the relative expression of the three GABR-related genes was detected in *M. persicae* exposed to the LC₃₀, LC₅₀, and LC₇₀ of betulin for 48 h. The relative expression of all three genes decreased gradually as the concentration of betulin increased (**Fig. 3B-D**). After exposure to the LC₃₀ of betulin for 48 h, the relative expression of *MpGABR*, *MpGABRAP* and *MpGABRB* decreased by 82.91%, 10.53%, and 11.49%, respectively. These results indicated that *MpGABR* was most sensitive to betulin.

Furthermore, after these three genes were silenced via RNAi (**Fig. 3H**), the expression levels of *MpGABR*, *MpGABRAP* and *MpGABRB* in *M. persicae* decreased by 64.57%, 63.28%, and 67.33%, respectively, compared with those in the control group (DEPC-treated water) (**Fig. 3E-G**). After exposure to the LC₅₀ of betulin for 48 h, the mortality of *M. persicae* with *MpGABR* silenced markedly increased, by 30.44%, compared with that in the control group, whereas the mortality of *M. persicae* in the *MpGABRAP*- and *MpGABRB*-silenced groups did not significantly differ from that in the control group (**Fig. 3I**). Additionally, western blotting analysis revealed that *MpGABR* protein expression decreased by 69.91% after RNAi (Fig. S1A, B). These results further implied that *MpGABR* might be a target for betulin against *M. persicae*.

3.5. Phylogenetic analysis of GABRs in insects

Furthermore, phylogenetic analysis of GABRs in insects was performed using the amino acid sequences of 71 GABR proteins from Hemiptera, Diptera, Lepidoptera, Thysanoptera, Hymenoptera, and Coleoptera. *MpGABR* (GenBank number: XM022318019.1) was found to be genetically closely related to CAI6365831.1 from *Macrosiphum euphorbiae* (**Fig. 4** and Table S6), indicating that *MpGABR* may be functionally analogous to CAI6365831.1.

3.6. Responses of *MpGABR* to betulin

To investigate the effect of betulin on the *MpGABR* protein, *M. persicae* was exposed to the LC₃₀, LC₅₀, and LC₇₀ of betulin for 48 h. As the concentration of betulin increased, the *MpGABR* protein content in *M. persicae* gradually decreased (**Fig. 5A and B**). Compared with that in the CK group, the *MpGABR* protein content in *M. persicae* exposed to the LC₃₀, LC₅₀, and LC₇₀ of betulin decreased by 33.68%, 44.89%, and 76.89%, respectively.

Moreover, the WT (wild type) *MpGABR* was expressed in *Escherichia coli* (**Fig. 5C**). The binding affinity of betulin with WT *MpGABR* was further measured using MST. The K_d value was 2.24 μ M (**Fig. 5D**), suggesting that betulin could bind to *MpGABR*. Additionally, voltage-clamp-based electrophysiological recording demonstrated that *MpGABR* strongly responded to betulin (**Fig. 5E**). In the presence of only GABA, a fast inward current was generated. As the concentration of

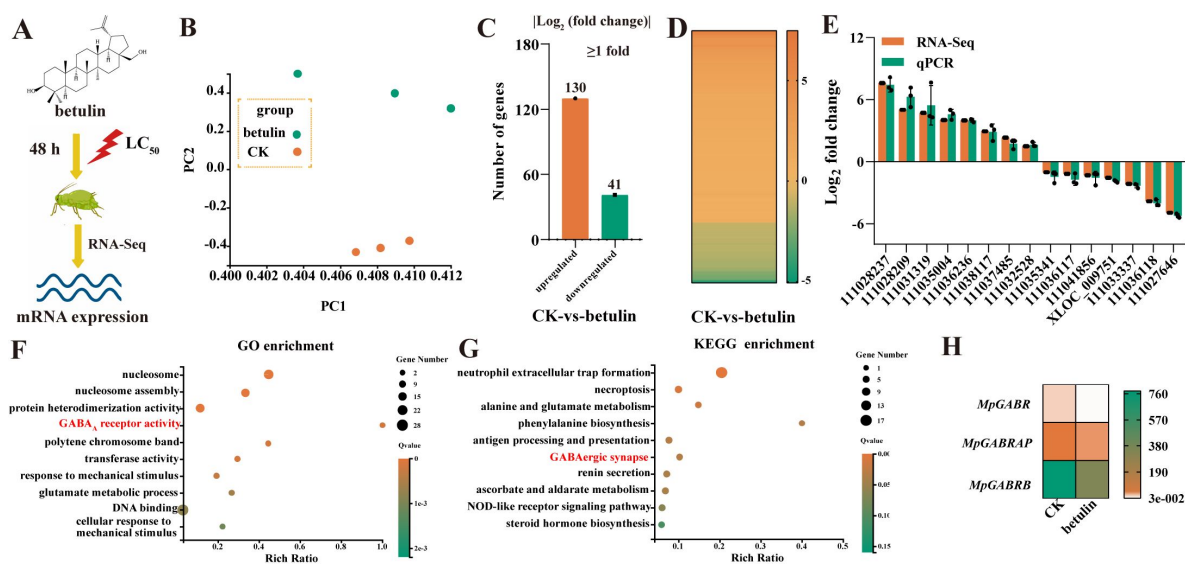


Figure 2.

RNA-Seq analysis revealed candidate targets for betulin against *M. persicae*.

(A) Diagram of the candidate targets for betulin against aphids as determined by RNA-Seq. Water containing 0.1% (v/v) Tween-80 and 3% (v/v) acetone was used as the control treatment (CK). (B) Principal coordinates analysis (PCA) analysis differentially expressed genes in RNA-Seq. (C-D) Distribution (C) and heatmap (D) of the significantly differentially expressed genes (DEGs) in *M. persicae*. (E) qPCR validation of 15 DEGs identified by RNA-Seq. (F-G) Top 10 enriched GO (F) and KEGG (G) pathways of the DEGs. The "rich ratio" was defined as the ratio of the number of DEGs enriched in the pathway to the total number of genes enriched in the same pathway. (H) Heatmap of the expression level of the GABR genes identified by RNA-Seq. *MpGABR*, encoding GABA_A receptor; *MpGABRAP*, encoding GABA_A receptor associated protein; *MpGABRB*, encoding GABA_A receptor β subunit.

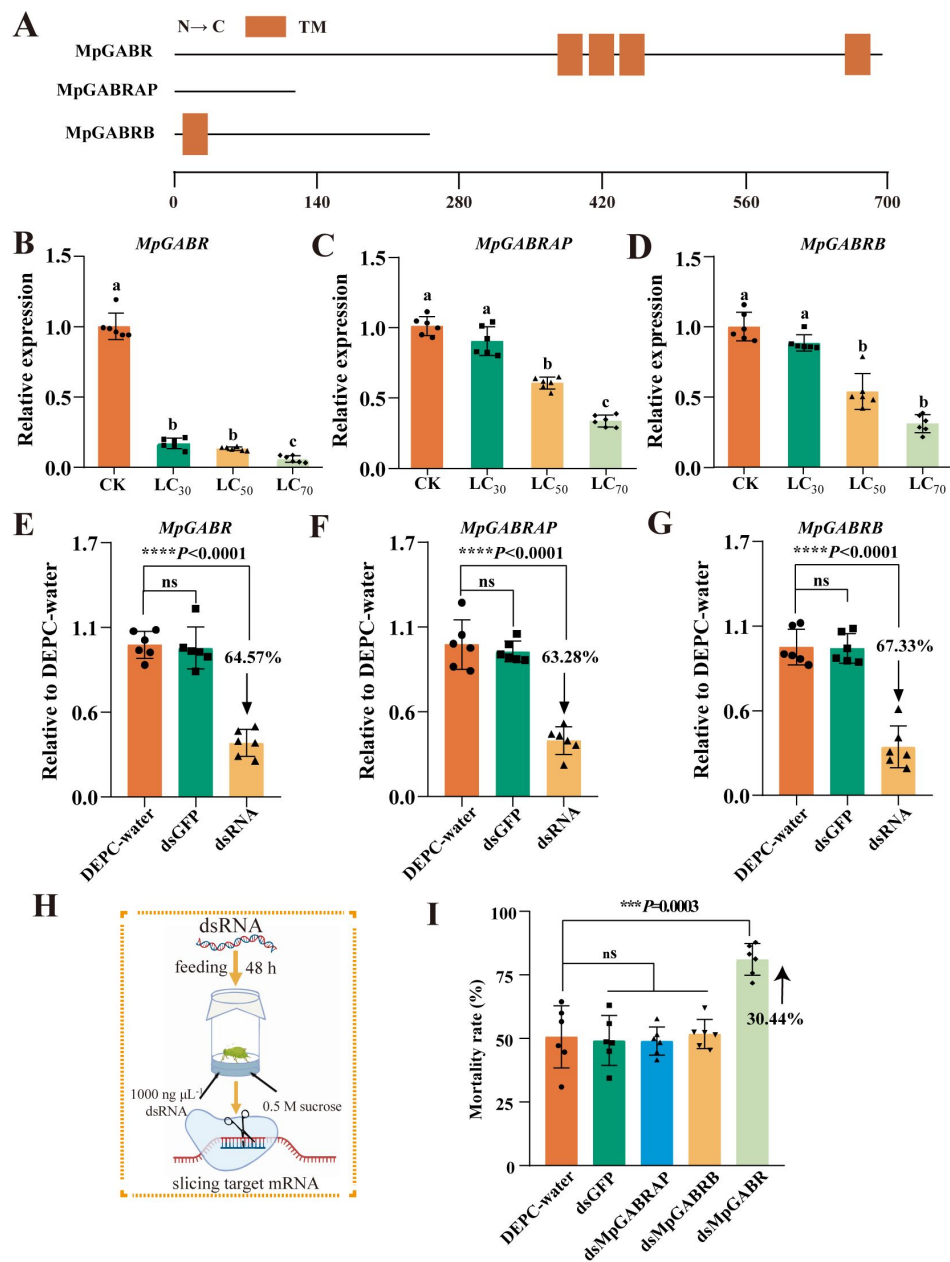


Figure 3.

MpGABR expression in aphids was significantly inhibited by treatment with betulin.

(A) Schematic drawing of MpGABR, MpGABRAP and MpGABRB. TM, transmembrane helices. (B-D) qPCR expression analysis of *MpGABR* (B), *MpGABRAP* (C) and *MpGABRB* (D) transcripts in *M. persicae* exposed to LC₃₀, LC₅₀, and LC₇₀ of betulin for 48 h. The bars represent the average (± SD). Different letters above the error bars indicate significant difference (ANOVA, Tukey's test, $P < 0.05$). (E-G) qPCR expression analysis of *MpGABR* (E), *MpGABRAP* (F) and *MpGABRB* (G) after RNAi at 48 h post-dsRNA feeding relative to the expression levels after DEPC-water treatment. (H) Schematic drawing of the RNAi assay in *M. persicae*. (I) Mortality of aphids exposed to the LC₅₀ of betulin for 48 h after RNAi. An asterisk (*) on the error bar indicates a significant difference between the treatment and group CK according to t tests, *** $P < 0.001$, **** $P < 0.0001$. ns, not significant.

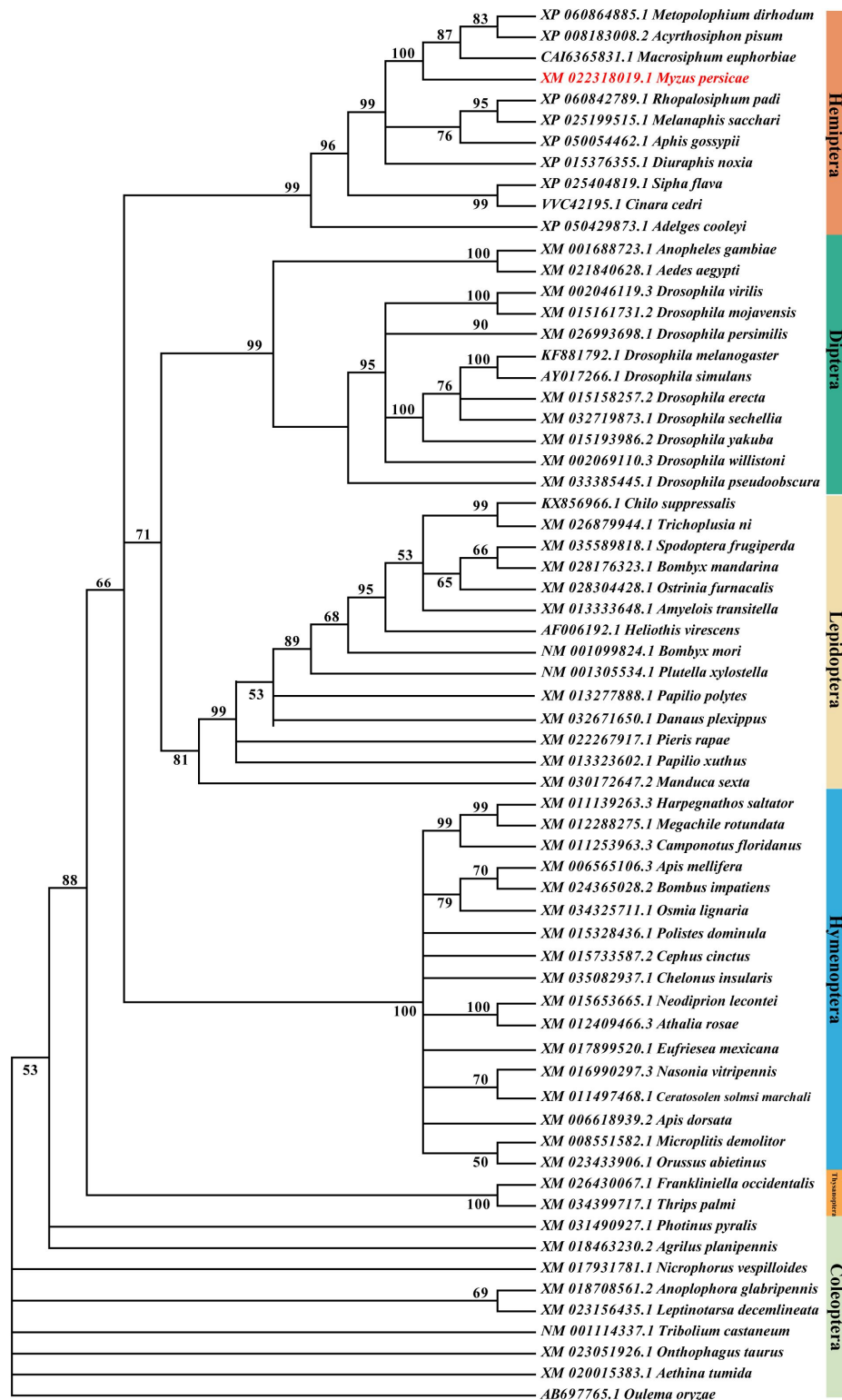


Figure 4.

Phylogenetic analysis of GABR in insects.

Maximum likelihood based phylogenetic tree of GABR from six orders: Hemiptera, Diptera, Lepidoptera, Thysanoptera, Hymenoptera, and Coleoptera. Protein sequence alignment was performed using MUSCLE in MEGA X. The sequences used for constructing the tree were listed in Table S6.

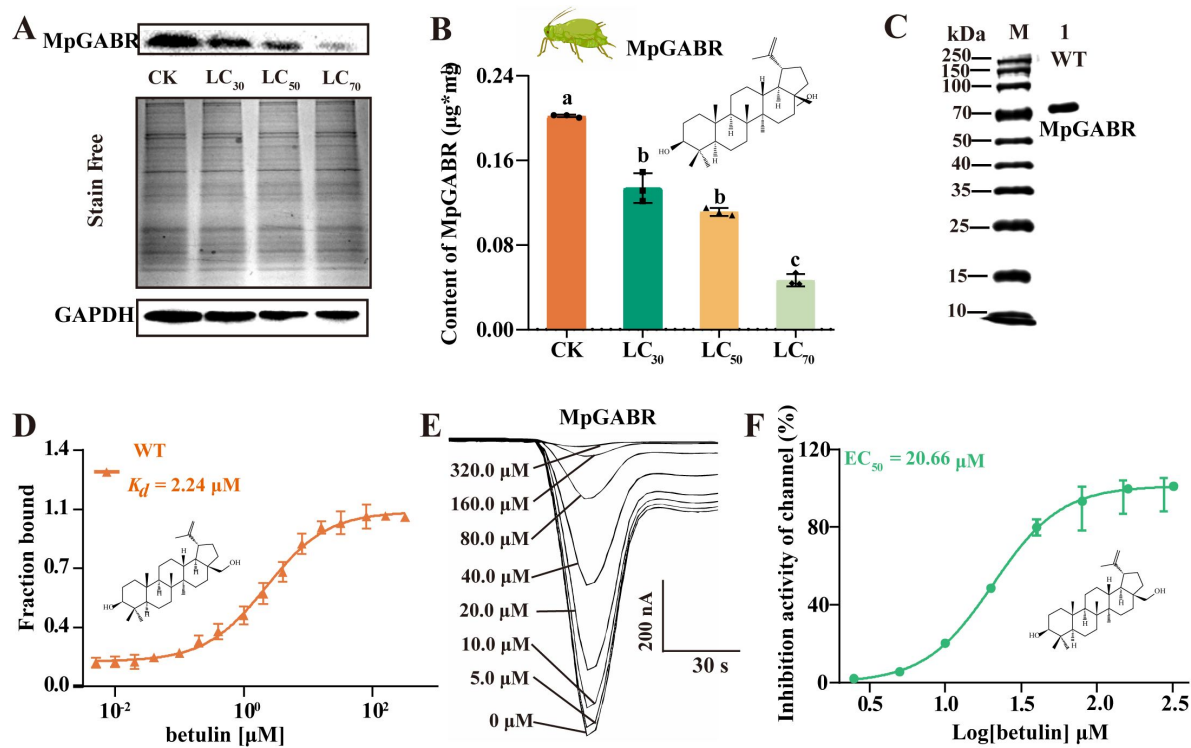


Figure 5.

The interaction of betulin with MpGABR protein.

(A, B) Western blotting analysis of MpGABR protein after betulin treatment for 48 h at three different concentrations (LC₃₀, LC₅₀ and LC₇₀). The bars represent the average ($\pm$ SD). Different letters above the error bars indicate significant difference (ANOVA, Tukey's test, $P < 0.05$). (C) Expression of recombinant wild type-aphid MpGABR (1, WT) protein by *Escherichia coli*. (D) Quantification of the binding affinity of betulin with wild type-aphid MpGABR using MST. Current responses (E) and inhibition activity (F) of MpGABR induced by different concentration of betulin in the presence of GABA. 0 μM MpGABR indicates the presence of only GABA.

added betulin increased, the current and channel activity of MpGABR decreased correspondingly, with an EC_{50} value of 20.66 μM (Fig. 5E and F). These results indicated that betulin acted as an inhibitor of MpGABR.

3.5 Key sites for the binding of betulin to MpGABR

To further study the binding mode of betulin in the active pocket of MpGABR, molecular docking was performed to explore the structure–function relationships between betulin and MpGABR. The docking results revealed that the predicted binding energy between betulin and MpGABR was $-6.38 \text{ kcal}\cdot\text{mol}^{-1}$ (Table S7), verifying that betulin could act as a specific ligand for MpGABR. The three-dimensional binding pattern of betulin with MpGABR indicated that the four key amino acid residues (ARG224, ALA226, PHE227 and THR228) interacted with betulin in the MpGABR binding pocket (Fig. 6A). Among the four key amino acid residues, ALA226 and THR228 interacted with betulin via hydrogen bonding in the MpGABR binding pocket (Table S7). Betulin generated nonconventional H-bonds (C...H) with ALA226 (3.31 Å) and conventional H-bonds (C–O H) with THR228 (2.16 Å). Furthermore, sequence alignment of the key amino acid residues of GABR in different species from Hemiptera, Diptera, Lepidoptera, Hymenoptera, Thysanoptera, and Coleoptera indicated that only THR228 was conserved in Hemiptera (Fig. 6B), implying that THR228 may be an essential specific site for the action of betulin on MpGABR in aphids. Mutants of aphid MpGABR were subsequently constructed, including R224A (ARG224 replaced by ALA), A226T (ALA226 replaced by THR), F227Y (PHE227 replaced by TYR, similar to the site in *Drosophila*), and T228R (THR228 replaced by ARG, similar to the site in *Drosophila*) (Fig. 6C). The binding affinity of betulin with these mutants was measured using MST. The results revealed that the K_d values of R224A, A226T, F227Y, T228R, and WT were 2.31, 2.27, 2.25, 5321, and 2.24 μM , respectively (Fig. 6D and E, Table S8). This result suggested that the mutation at the THR228 site, with the highest K_d value, markedly reduced the binding ability of betulin to MpGABR, implying that the THR228 site may be an essential specific site for the binding of betulin to MpGABR.

3.6 Betulin binds specifically to MpGABR via THR228

To further prove that THR228 is the specific binding site for betulin in MpGABR, the WT and mutant (R122T, equivalent to THR228 in MpGABR) *Drosophila* DmGABR proteins were expressed in *E. coli* and extracted (Fig. 7A and D). The binding affinities of betulin and pymetrozine with DmGABR^{WT} and DmGABR^{R122T} were evaluated using MST. As shown in Fig. 7B, betulin was able to bind to DmGABR^{R122T} ($K_d = 342.4 \mu\text{M}$) but not to DmGABR^{WT}. However, pymetrozine exhibited no binding affinity for DmGABR^{R122T} or DmGABR^{WT} (Fig. 7C). Moreover, to generate *Drosophila* mutant individuals carrying DmGABR^{R122T}, the CRISPR/Cas9 genomic editing strategy was used. After exposure to different concentrations of betulin, the mortality rate of DmGABR^{R122T} *Drosophila* was significantly greater than that of DmGABR^{WT} *Drosophila* (Fig. 7E). In addition, the LD_{50} value of betulin against DmGABR^{R122T} *Drosophila* at 72 h was 27.19 $\mu\text{g}\cdot\text{fly}^{-1}$, whereas the LD_{50} value of betulin against DmGABR^{WT} *Drosophila* could not be calculated because the lethal concentration exceeded 1000 $\mu\text{g}\cdot\text{fly}^{-1}$ (Table S9). Additionally, there was no significant difference in mortality between DmGABR^{WT} and DmGABR^{R122T} *Drosophila* after exposure to pymetrozine (Fig. 7F). The LD_{50} values of pymetrozine against DmGABR^{WT} and DmGABR^{R122T} *Drosophila* at 72 h were 0.23 and 0.24 $\mu\text{g}\cdot\text{fly}^{-1}$, respectively (Table S9). Moreover, THR228 is located at the neurotransmitter-gated ion-channel ligand-binding domain in MpGABR (Fig. S2), implying that betulin may play a role as a competitive antagonist of MpGABR. Taken together, these findings suggested that betulin bound specifically to MpGABR via THR228, resulting in the death of *M. persicae*.

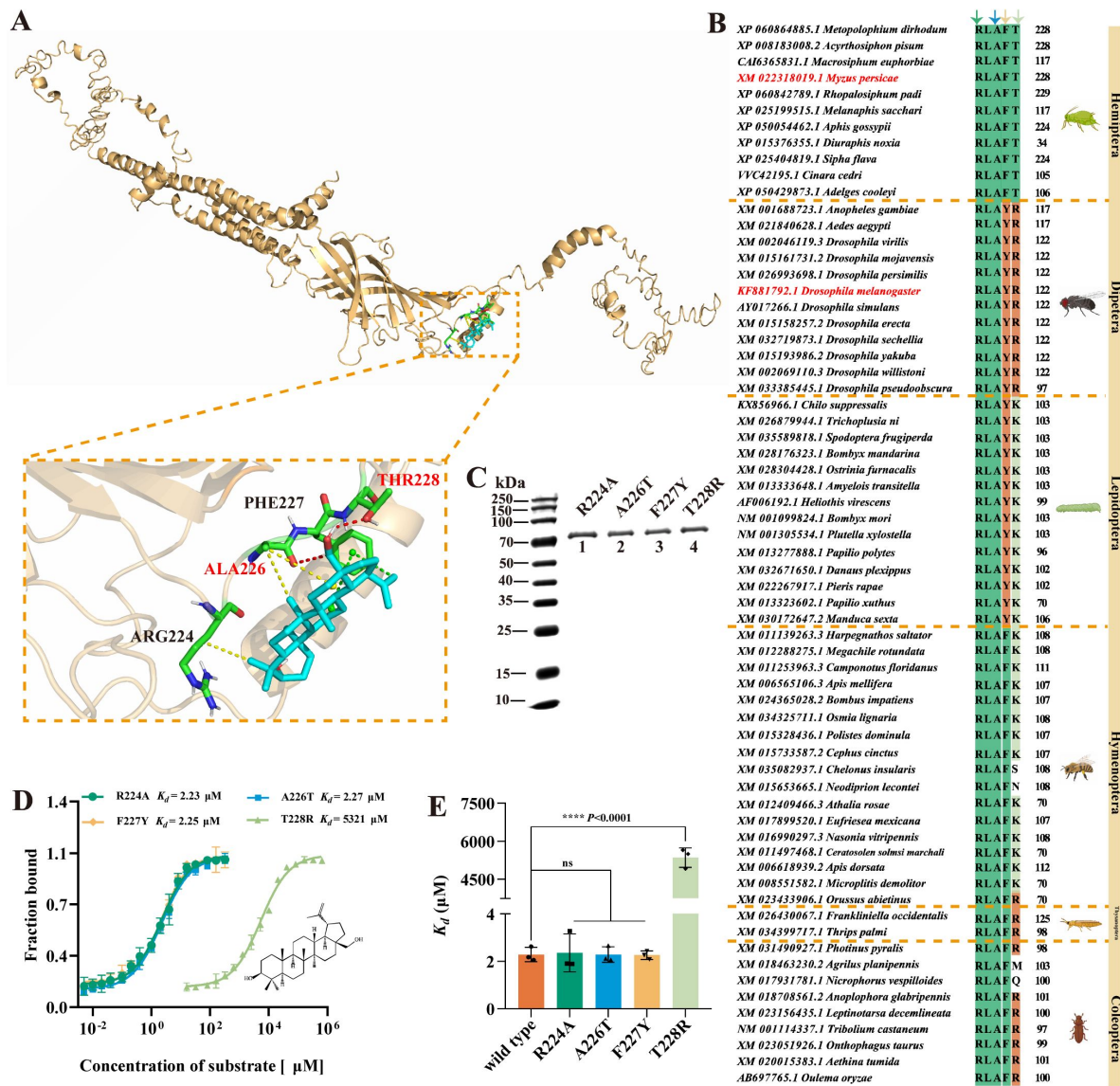


Figure 6.

Molecular docking, binding site and inhibitory effect of betulin on MpGABR.

(A) Best conformations of betulin docked to the binding pocket of MpGABR in aphids. An enlarged view of the betulin binding sites in MpGABR is indicated by a dashed frame. Potential Pi-alkyl (green), Alkyl (yellow), and hydrogen bond (red) interactions are indicated by dashed lines. (B) Sequence alignment of the key amino acids bound to betulin. The conserved residues among the different species in GABR are shown in orange, cyan, and light green. (C) Expression of the recombinant mutation type-aphid MpGABR by *Escherichia coli*. Lane 1: R224A, Lane 2: A226T, Lane 3: F227Y, Lane 4: T228R. (D, E) Quantification of the binding affinity of betulin with wild-type and mutant type-aphid MpGABR using MST. The bars represent the average ($\pm$ SD) values. * indicates a significant difference (Student's *t* test, *****P* < 0.0001).

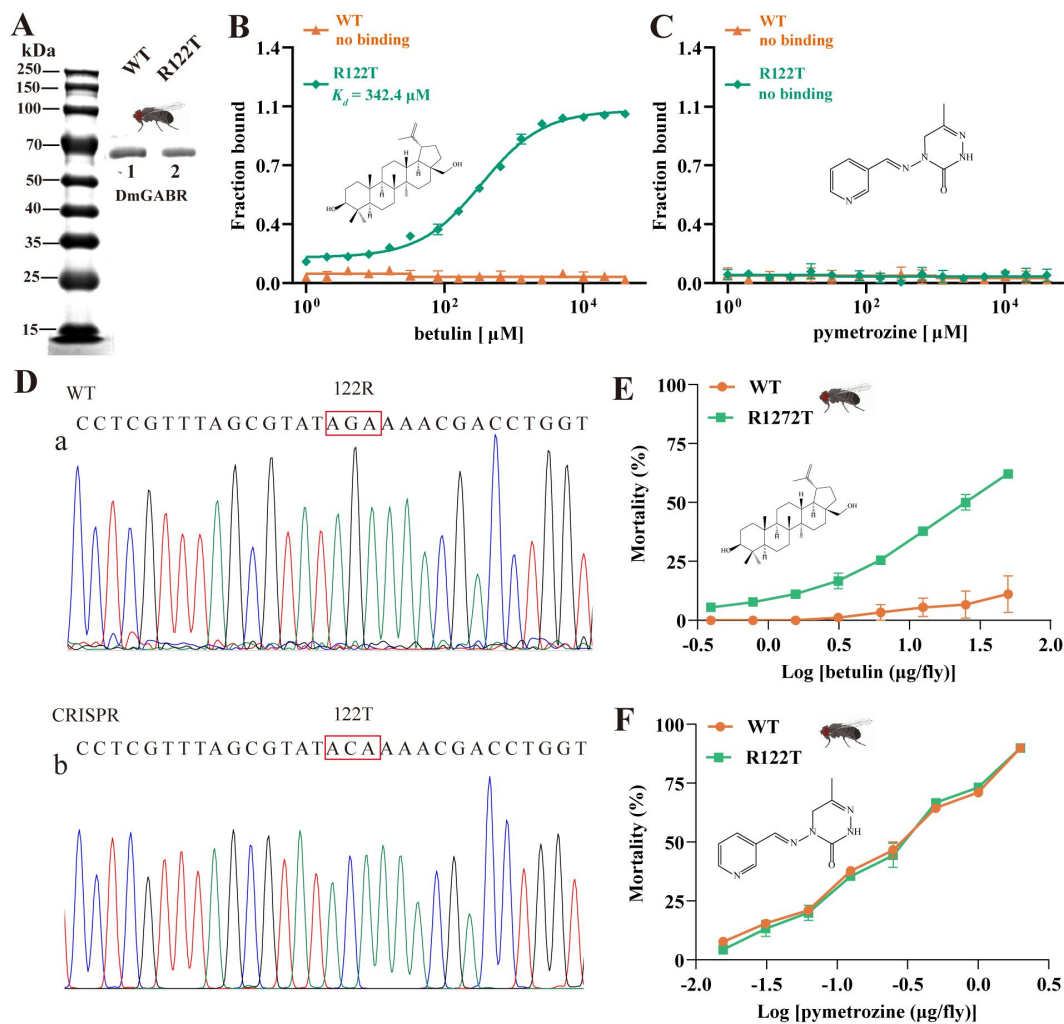


Figure 7.

Gene editing in *Drosophila* validated the species-specific binding site of betulin.

(A) Expression of the recombinant wild type (WT) and mutation type (R122T) of DmGABR (*Drosophila melanogaster*) by *E. coli*. Lane 1: wild type, Lane 2: R122T. (B, C) Quantification of the binding affinity of betulin (B) and pymetrozine (C) with WT and R122T of DmGABR using MST. (D) Sanger sequencing of the *DmGABR* gene in flies. Direct sequencing chromatograms of PCR products amplified from a fragment of gDNA flanking the WT (a), and introduced R122T (b) mutant flies. (E-F) Toxicity curves of betulin (E), and pymetrozine (F) against WT, and the DmGABR^{R122T} of the Cas9 *Drosophila*.

4. Discussion

Although chemical pesticides are effective in pest control, the long-term unreasonable application of these substances has led the emergence of resistant pests, environmental deterioration, and deleterious effects on nontarget organisms, including humans, raising widespread and intense concerns.³⁰ These challenges have made it urgent for the development of alternative strategies for pest control. Recently, the utilization of plant secondary metabolites as insecticides has become increasingly popular as an eco-friendly and biocontrol approach.^{31–33} Our previous study revealed that *P. davidiana*, a wild relative of cultivated peach, strongly resists *M. persicae* by accumulating high contents of betulin.^{3, 5} In addition, betulin, a lupane-type triterpene, possesses potent insecticidal activity and is a promising substance for the development of novel insecticides for aphid control. In this study, the insecticidal effect of betulin was further evaluated by comparing the control efficacy of betulin with that of pymetrozine against aphids in greenhouses and fields (**Fig. 1A–G**). These results indicate that betulin has a similar control effect to pyridone and has immense potential for development as a plant-derived insecticide.

Terpenes are a diverse group of plant secondary metabolites that can increase the resistance of plants to insect herbivores through direct³⁴ and indirect³⁵ defense mechanisms. In direct defense against herbivores, triterpenes play important roles in diverse biological activities, including antiparasitic, insecticidal, and antifeedant activities.^{36, 37} Azadirachtin, a tetracyclic triterpenoid compound isolated from the Indian neem tree (*Azadirachta indica*), is one of the most prominent commercial biopesticides, exhibiting strong insect antifeedant properties as well as growth- and reproduction-regulating effects.^{38, 39} Besides, triterpene glycoside compounds play crucial roles in the defense of tobacco (*N. attenuata*) against tobacco hornworm (*Manduca sexta*) larvae.⁴⁰ Although betulin exhibits various pharmacological activities,^{14–17} reports on its insecticidal activity are limited. Encouragingly, the effects of betulin and its derivatives on pests have attracted increasing attention. Betulinic acid and its derivatives showed larvicidal activity against *Aedes aegypti* larvae.⁴¹ Betulin-cinnamic acid-related hybrid compound 5b exhibited strong aphicidal activity, and compound 2l could destroy the ultrastructure of midgut cells and significantly inhibit the activity of α -amylase in diamondback moth (*Plutella xylostella* L.) larvae.⁴² Our previous studies also indicated that betulin possesses potent insecticidal activity and is a key endogenous secondary metabolite related to the defense of peach against *M. persicae*.⁵ Elucidating the insecticidal mechanism of betulin against aphids will provide a basis for the development of novel aphicides and sustainable strategies for aphid control.

GABA receptors have been confirmed to be targets of terpenoids that impair insect neuronal function in herbivores.¹⁹ GABRs are heteropentameric ligand-gated ion channels in the central nervous system that conduct chloride and bicarbonate ions. These receptors are targets of numerous drugs for the treatment of neuropsychiatric disorders.²⁰ A variety of terpenoids act as positive allosteric modulators or noncompetitive antagonists of GABRs, such as diterpenoids (isopimaric acid and miltirone), sesquiterpenoids (picrotoxin, bilobalide and ginkgolides) and monoterpenoids (α -thujone and thymol).¹⁹ In this study, both GO and KEGG enrichment analyses revealed that the DEGs identified by RNA-seq were enriched in GABAergic signaling-related pathways (**Fig. 2F and G**). Additionally, the expression of the DEGs related to GABRs, particularly *MpGABR* (Table S4), in the betulin group was significantly lower than that in the CK group (**Fig. 2H**). Besides, the relative expression of *MpGABR*, *MpGABRAP* and *MpGABRB* decreased gradually after *M. persicae* was exposed to the LC₃₀, LC₅₀, and LC₇₀ of betulin for 48 h (**Fig. 3B–D**). Among them, *MpGABR* was the most sensitive to betulin, and its expression was reduced by 82.91% after exposure to the LC₃₀ of betulin for 48 h. Furthermore, compared with the control group, the *M. persicae* group with *MpGABR* silenced by RNAi presented a significant

increase in mortality, by 30.44%, after 48 h of exposure to the LC_{50} of betulin (**Fig. 3I**). Collectively, these results suggest that betulin may have insecticidal effects on aphids by inhibiting *MpGABR* expression.

Additionally, betulin has been reported to be able to bind to GABRs.⁴³ We further investigated the interaction of betulin with the *MpGABR* protein. The MST assay revealed that betulin was able to bind to *MpGABR* ($K_d = 2.24 \mu M$) (**Fig. 5D**), which is consistent with previous findings showing that betulin binds to GABA receptors in mouse brains *in vitro*.⁴⁴ Voltage-clamp-based electrophysiological recordings indicated that betulin acted as an inhibitor ($EC_{50} = 20.66 \mu M$) for *MpGABR* (**Fig. 5E and F**). Subsequent molecular docking analysis suggested that four key amino acid residues (ARG224, ALA226, PHE227 and THR228) interact with betulin in the *MpGABR* binding pocket (**Fig. 6A**), among which only ALA226 and THR228 interact with betulin via hydrogen bonding (Table S7). The results of the sequence alignment revealed that only THR228 was conserved in Hemiptera (**Fig. 6B**). Furthermore, evaluation of the ability of betulin to bind to *MpGABR* mutants with mutations at those four sites revealed that the ability of betulin to bind to T228R was significantly weaker than its ability to bind to the WT (**Fig. 6D and E**, Table S8), indicating that THR228 is an essential specific site for the binding of betulin to *MpGABR*. Moreover, to further prove that THR228 is the specific binding site for betulin in *MpGABR*, the binding affinities of betulin with the WT and mutant (R122T, equivalent to THR228 in *MpGABR*) *Drosophila* DmGABR proteins were assessed using MST. The results showed that betulin was able to bind to DmGABR^{R122T} ($K_d = 342.4 \mu M$) but not DmGABR^{WT} (**Fig. 7B**). Additionally, after exposure to different concentrations of betulin, the mortality rate of DmGABR^{R122T} *Drosophila* was significantly greater than that of DmGABR^{WT} *Drosophila* (**Fig. 7E** and Table S9). Similarly, a previous study indicated that the R122G amino acid site substitution, generated via RNA editing, affects the sensitivity of *Drosophila* to fipronil.⁴⁵ Together, these findings suggest that betulin binds specifically to *MpGABR* via THR228, acting as an inhibitor of *MpGABR* and causing aphid death.

GABRs belong to the cysteine loop (Cys)-loop superfamily of neurotransmitter receptors and are essential heteropentameric ligand-gated ion channels in the central nervous system. After GABA binds to the extracellular Cys-loop of a GABR,⁴⁶ the GABR is activated and opens up its central pore to allow chloride ions to pass through, thereby hyperpolarizing neurons and attenuating excitatory neurotransmission.⁴⁷ In insects, GABRs play an important role in circadian rhythms,⁴⁸ sleep,⁴⁹ movement,⁵⁰ olfactory memory,⁵¹ and so on. Additionally, GABR is a critical target for a variety of insecticides and ectoparasiticides. Insecticides targeting GABRs are categorized into noncompetitive antagonists (NCAs) and competitive antagonists (CAs) according to their different binding sites. NCAs block chloride channels in nerve cells by interacting with amino acid residues of the GABA-gated chloride channel, causing a conformational change in the receptor, which interferes with the normal function of the central nervous system and ultimately leads to insect death.^{19, 52} A series of first- and second-generation NCA insecticides have been successfully developed. Picrotoxinin, isolated from *Anamirta cocculus* fruit, is the oldest natural NCA and is toxic to houseflies.⁵³ Among the first-generation NCAs, polychlorocycloalkanes, including dieldrin and lindane, were commonly used as pesticides in the middle and late 20th centuries.^{54, 55} As second-generation NCAs, fipronil and its derivatives are commercially available phenylpyrazole insecticides that have been widely used for agricultural pest control.^{56, 57} Additionally, CAs bind to orthogonal binding sites, such as the GABA recognition site in the extracellular region, competitively inhibiting the binding of GABA to its receptor, leading to toxic effects in insects. Given the lack of interference from existing insect resistance mechanisms, CAs hold promise for the development of efficient new insecticides. Novel 1, 6-dihydro-6-iminopyridazine-derived insecticides, as CAs, have insecticidal properties against common cutworms and houseflies.⁵⁸ Besides, nootkatone, a sesquiterpenoid, also acts as a CA to induce insect mortality.⁵⁹ Fluralaner, an isooxazoline ectoparasiticide, inhibits both parasites and *Aedes aegypti* by acting as a CA.^{60, 61} Our results revealed that the binding site (THR228) for betulin in *MpGABR* was located in the extracellular neurotransmitter-gated ion-channel ligand-binding

domain (Fig. S2), implying that betulin acts as a CA of MpGABR. Our work may have provided a novel target for developing insecticides. In this study, betulin, on the one hand, inhibited the expression of *MpGABR* and, on the other hand, specifically bound to MpGABR through THR228, acting as an inhibitor of MpGABR and causing aphid death (Fig. 8).

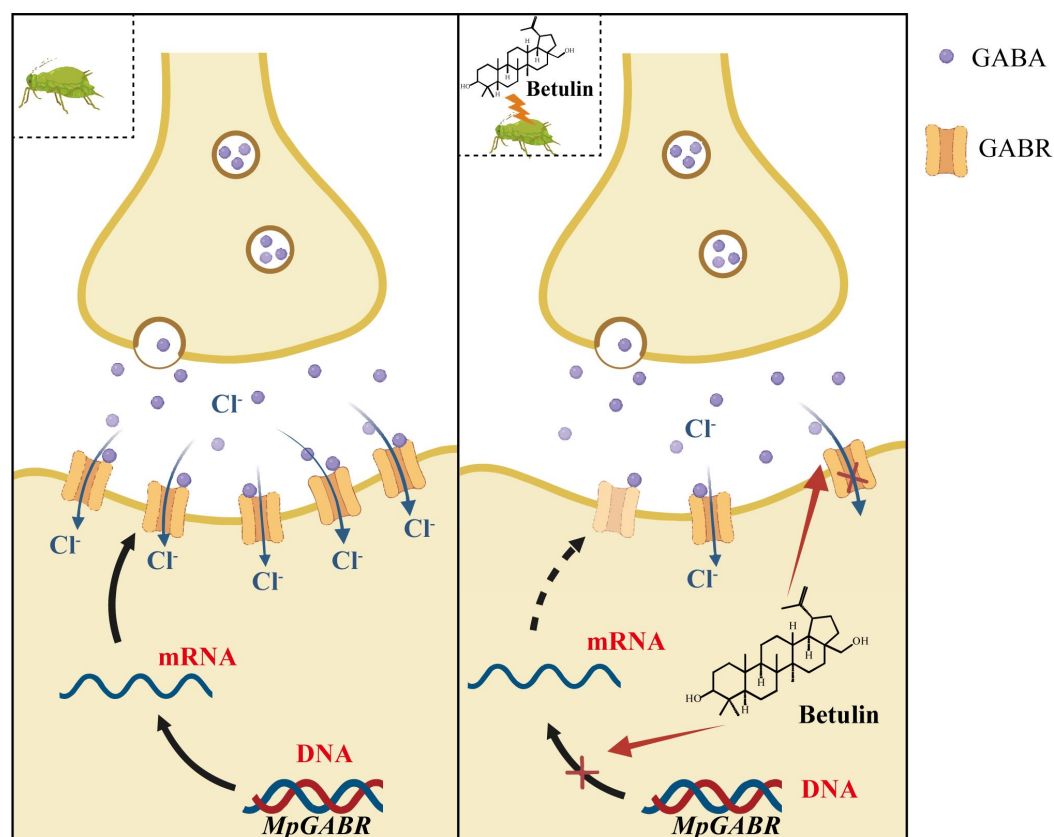


Figure 8.

Proposed model for the mechanism of action against *Myzus persicae* by targeting GABA_A receptors (GABR).

After exposure to betulin, the expression of *MpGABR* was inhibited the level of MpGABR protein is decreased, resulting in a decrease in the channel of chloride ion influx. Besides, betulin directly and specifically binds to the amino acid residue THR228 of MpGABR, thereby disabling it.

5. Conclusion

Betulin, a key metabolite in the aphid-resistant wild peach *P. davidiana*, possesses potent aphicidal effects on *M. persicae*. This study confirmed that betulin exhibited excellent control efficacy against *M. persicae* in both greenhouse and field experiments. RNA-seq, qRT-PCR and western blotting assays revealed that betulin significantly inhibited the expression of *MpGABR* in aphids. In addition, RNAi-mediated silencing of *MpGABR* markedly increased the sensitivity of aphids to betulin. Moreover, MST and voltage-clamp-based electrophysiological recording assays indicated that betulin was able to bind to MpGABR ($K_d = 2.24 \mu\text{M}$) and acted as an inhibitor ($\text{EC}_{50} = 20.66 \mu\text{M}$) of MpGABR. Molecular docking analyses suggested that the amino acid residue THR228, which is highly conserved in Hemiptera, might be a critical specific binding site for betulin in MpGABR.

Mutagenesis and genome editing assays revealed that betulin bound specifically to this amino acid residue in aphids but not in *Drosophila*, resulting in aphid death. Collectively, the results suggest that the aphicidal effects of betulin on aphids occur in a two-pronged manner: on the one hand, betulin inhibits *MpGABR* expression; on the other hand, it specifically binds to *MpGABR* via THR228 and acts as an inhibitor of *MpGABR*. Elucidating the insecticidal mechanism of betulin against aphids will provide a basis for the development of novel insecticides and sustainable strategies for aphid control.

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Notes

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Additional files

Supplementary Materials. Protein expression of MpGABR after RNAi (Figure S1); Betulin binding site on the domain of MpGABR (Figure S2); LC_{50} values of betulin and pymetrozine against *M. persicae* at 48 h (Table S1); Primers used for cloning, RNAi, heterologous expression, and qRT-PCR (Table S2); Oligonucleotide sequences for PCR, the sgRNA synthesis template and the fragment of donor DNA for homology-directed repair (HDR) (Table S3); Screening of genes significantly differentially expressed in *M. persicae* in response to betulin at 48 h post treatment (Table S4); Complete sequence information for the GABA_A receptor gene of *M. persicae* (Table S5); Sequences and relevant information for phylogenetic analysis of GABA_A receptor (Table S6); Binding energy and nonbonding interactions between betulin and GABA_A receptor (Table S7); Parameters of dose-response curves from microscale thermophoresis experiments (Table S8); LD_{50} values of betulin and pymetrozine against *D. melanogaster* at 72 h (Table S9). [↗](#)

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Reviewer #1 (Public review):

Wang, Junxiu et al. investigated the underlying molecular mechanisms of the insecticidal activity of betulin against the peach aphid, *Myzus persicae*. There are two important findings described in this manuscript: (a) betulin inhibits the gene expression of GABA receptor in the aphid, and (b) betulin binds to the GABA receptor protein, acting as an inhibitor. The first finding is supported by RNA-Seq and RNAi, and the second one is convinced with MST and electrophysiological assays. Further investigations on the betulin binding site on the receptor protein provided a fundamental discovery that T228 is the key amino acid residue for its affinity, thereby acting as an inhibitor, backed up by site-directed mutagenesis of the heterologously-expressed receptor in *E. coli* and by CRISPR-genome editing in *Drosophila*.

Although the manuscript does have strengths in principle, the weaknesses do exist: the manuscript would benefit from more comprehensive analyses to fully support its key claims in the manuscript. In particular:

(1) The Western blotting results in Figure 5A & B appear to support the claim that betulin inhibits GABR gene expression (L26), as a decrease in target protein levels is often indicative of suppressed gene expression. The result description for Figure 5A & B is found in L312-L316, within Section 3.6 ("Responses of MpGABR to betulin"), where MST and voltage-clamp assays are also presented. It seems the observed decrease in MpGABR protein content is due to gene downregulation, rather than a direct receptor protein-betulin interaction. However, this interpretation lacks discussion or analysis in either the corresponding results section or the Discussion. In contrast, Figures 5C-F are specifically designed to illustrate protein-betulin interactions. Presenting Figure 5A & B alongside these panels might lead to confusion, as they support distinct claims (gene expression vs. protein binding/inhibition). Therefore, I recommend moving Figure 5A & B either to the end of Figure 3 or to a separate figure altogether to improve clarity and logical flow. A minor point in the Western blotting experiment is that although GAPDH was used as a reference protein, there is no explanation in the corresponding M&M section.

(2) The description of the electrophysiological recording experiment is unclear regarding the use of GABA. I didn't realize that GABA, the true ligand of the GABA receptor, was used in this inhibition experiment until I reached the Results section (L321), which states, "In the presence of only GABA, a fast inward current was generated." Crucially, no details are provided on the experiment itself, including how GABA was applied (e.g., concentration, duration, whether GABA was treated, followed by betulin, or vice versa). This information is essential for reproducibility. Please ensure these details are thoroughly described in the corresponding M&M section.

(3) The phylogenetic analysis, particularly concerning Figures 4 and 6B, needs significant attention for clarity and representativeness. First, your claim that MpGABR is only closely related to CAI6365831.1 (L305-L310) is inconsistent with the provided phylogenetic tree, which shows MpGABR as equally close to *Metopolophium dirhodum* (XP_060864885.1) and *Acyrtosiphon pisum* (XP_008183008.2). Therefore, singling out only *Macrosiphum euphorbiae* (CAI6365831.1) is not supported by the data. Second, the representation of various insect orders is insufficient. All 11 sequences in the Hemiptera category (in both Figure 4 and Figure 6B) are exclusively from the Aphididae family. This small subset cannot represent the highly diverse Order Hemiptera. Consequently, statements like "only THR228 was conserved in Hemiptera" (L338), "The results of the sequence alignment revealed that only THR228 was conserved in Hemiptera" (L430), or "THR228... is highly conserved in Hemiptera" (L486) are not adequately supported. Third, similar concerns apply to the Diptera

order, which includes 10 *Drosophila* and 2 mosquito samples (not diverse or representative enough), and likely to other orders as well. Thereby, the Figure 6B alignment should be revised accordingly to reflect a more accurate representation or to clarify the scope of the analysis. Fourth, there's a discrepancy in the phylogenetic method used: the M&M section (L156) states that MEGA7, ClustalW, and the neighbor-joining method were used, while the Figure 4 caption mentions that MEGA X, MUSCLE, and the Maximum likelihood method were employed. This inconsistency needs to be clarified and made consistent throughout the manuscript. Fifth, I have significant concerns about the phylogenetic tree itself (Figure 4). A small glitch was observed at the *Danaus plexippus* node, which raises suspicion regarding potential manipulation after tree construction. More critically, the tree, especially within Coleoptera, does not appear to be clearly resolved. I am highly concerned about whether all included sequences are true GABR orthologs or if the dataset includes partial or related sequences that could distort the phylogeny. Finally, for Figure 6B, both protein (XP_) and nucleotide (XM_) sequences were mix used. I recommend using the protein sequences instead of nucleotide sequences in this figure panel, as protein sequences are more directly informative.

(4) The Discussion section requires significant revision to provide a more insightful and interpretative analysis of the results. Currently, much of the section primarily restates findings rather than offering deeper discussion. For instance, L409-L419 restate the results, followed by the short sentence "Collectively, these results suggest that betulin may have insecticidal effects on aphids by inhibiting MpGABR expression". It could be further expanded to make it beneficial to elaborate on proposed mechanisms by which gene expression might be suppressed, including any potential transcription factors involved. In contrast, while L422-L442 also initially summarize results, the subsequent paragraph (L445-L472) effectively discusses the potential mechanisms of inhibitory action and how mortality is triggered, which is a good model for other parts of the section. However, all the discussion ends up with a short statement, "implying that betulin acts as a CA of MpGABR" (L472), which appears to be a leap. The inference that betulin acts as a competitive antagonist (CA) is solely based on the location of its extracellular binding site, which does not exactly overlap with the GABA binding site. It needs stronger justification or actually requires further experimental validation. The authors should consider rephrasing this statement to acknowledge the need for additional studies to definitively confirm this mechanism of action.

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Reviewer #2 (Public review):

Summary:

This important study shows that betulin from wild peach trees disrupts neural signaling in aphids by targeting a conserved site in the insect GABA receptor. The authors present a nicely integrated set of molecular, physiological, and genetic experiments to establish the compound's species-specific mode of action. While the mechanistic evidence is solid, the manuscript would benefit from a broader discussion of evolutionary conservation and potential off-target ecological effects.

Strengths:

The main strengths of the study lie in its mechanistic clarity and experimental rigor. The identification of a betulin-binding single threonine residue was supported by (1) site-directed mutagenesis and (2) functional assays. These experiments strongly support the specificity of action. Furthermore, the use of comparative analyses between aphids and fruit flies demonstrates an important effort to explore species specificity, and the integration of quantitative data further enhances the robustness of the conclusions.

Weaknesses:

There are several important limitations that need to be addressed. The manuscript does not explore whether the observed sensitivity to betulin reflects a broadly conserved feature of GABA receptors across animal lineages or a more lineage-specific adaptation. This evolutionary context is crucial for understanding the broader significance of the findings.

In addition, while the compound's aphicidal effect is well established, the potential for off-target effects in non-target organisms - especially vertebrates - remains unaddressed, despite prior evidence that betulin interacts with mammalian GABA_A receptors. There is little discussion on the ecological or environmental safety of exogenous betulin application, such as persistence, degradation, or exposure risks.

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